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DEPARTMENT OF COMPUTER SCIENCE AND DESIGN

MINI PROJECT

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BONAFIDE CERTIFICATE

Certified that this project report titled “**Diabetes Prediction System Using Machine Learning**” is the Bonafide work of **HARISH BABU R (231701017)** who carried out the project work for the course **AI23331 - Fundamentals of Machine Learning** under my supervision.

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MINI PROJECT REPORT

DIABETES PREDICTION SYSTEM USING MACHINE LEARNING

1. INTRODUCTION

Diabetes is one of the most significant health problems affecting millions of people worldwide. Due to changes in lifestyle, food habits, stress, and lack of physical activity, the number of diabetic patients has increased rapidly in recent years. Diabetes occurs when the body is unable to properly regulate blood sugar levels because of insufficient insulin production or ineffective insulin usage. If diabetes is not detected at an early stage, it can lead to severe health complications such as heart disease, kidney failure, nerve damage, and vision loss.

Detecting diabetes at an early stage is important for:

1. Assisting doctors in making faster medical decisions
2. Reducing health risks through early treatment
3. Improving healthcare monitoring and patient awareness

This project uses Machine Learning techniques to predict diabetes using the publicly available Diabetes Dataset, which contains several medical attributes such as glucose level, blood pressure, insulin level, BMI, age, and pregnancies. The project also includes a clean and interactive User Interface (UI) that allows users to easily provide medical inputs and receive prediction results instantly.

2. OBJECTIVES

The objectives of this project are:

1. To predict diabetes using Machine Learning techniques
2. To preprocess and analyze the diabetes dataset for accurate prediction
3. To improve prediction performance using feature scaling techniques
4. To compare multiple machine learning models and identify the best-performing model
5. To evaluate the models using classification metrics such as Accuracy, Precision, Recall, and F1-Score
6. To develop a clean and interactive User Interface for real-time prediction

3. METHODOLOGY

The methodology of this project is structured into the following stages:

3.1. Data Acquisition

The dataset used in this project is the Diabetes Dataset available on Kaggle and other public machine learning repositories. The dataset contains medical information collected from patients and includes 9 features: Pregnancies, Glucose, Blood Pressure, Skin Thickness, Insulin, BMI, Diabetes Pedigree Function, Age, and the target variable Outcome. The Outcome column indicates whether the patient is diabetic (1) or non-diabetic (0). The dataset is widely used for healthcare prediction and machine learning research applications.

3.2. Preprocessing / Data Cleaning

To ensure better model performance and prediction reliability, the following preprocessing steps are performed:

1. Missing and invalid values are identified and handled appropriately
2. Duplicate records are checked and removed if necessary
3. Numerical features are standardized using StandardScaler
4. The dataset is verified for correct data types before model training

These preprocessing techniques improve the quality of the dataset and help the model learn effectively from medical data.

3.3. Feature Engineering

Feature engineering is performed to improve the prediction capability of the machine learning models:

1. Important medical parameters such as Glucose and BMI are analyzed carefully
2. Numerical features are scaled to maintain consistency in data distribution
3. Relevant healthcare attributes are selected to improve classification performance

These feature engineering techniques help the model identify hidden patterns associated with diabetes prediction.

3.4. Data Splitting

The dataset is divided into training and testing sets before model training. The training dataset is used to train the machine learning models, while the testing dataset is used to evaluate model performance on unseen data. Typically, 80% of the dataset is used for training and 20% is used for testing to ensure balanced and reliable evaluation.

3.5. Model Training

Two machine learning models are trained and compared:

1. **Regression** – A baseline linear classification model
2. **Random Forest Classifier** – An ensemble tree-based classification model

Both models are trained on the processed dataset and tested using the testing split to evaluate their performance for diabetes prediction.

3.6. Evaluation

The machine learning models are evaluated using the following metrics:

1. Accuracy – Measures the overall correctness of predictions
2. Precision – Measures how many predicted diabetic cases are correct
3. Recall – Measures how many actual diabetic cases are correctly identified
4. F1-Score – Harmonic mean of Precision and Recall
5. Confusion Matrix – Visual representation of prediction performance

These evaluation metrics help determine the effectiveness and reliability of the diabetes prediction system.

3.7. Threshold Tuning

By default, classifiers use a 0.5 probability threshold. For fraud detection, minimizing false negatives (missed frauds) is more important than avoiding false positives. The decision threshold is lowered to 0.3, which increases Recall and ensures more fraud cases are caught, even at a slight cost to Precision.

3.8. Real-Time Prediction Simulation

A real-time diabetes prediction function is implemented by integrating the trained machine learning model with the user interface. The system accepts patient medical data as input, processes the values using the trained model, and predicts whether the patient is diabetic or non-diabetic.

This simulates a real-world healthcare prediction application that can assist in early diabetes detection and medical support systems.

4. CODE

4.1. Import Libraries

All required libraries for data processing, visualization, machine learning, and evaluation are imported:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import streamlit as st

from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LogisticRegression
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import (
    accuracy_score,
    classification_report,
    confusion_matrix
)

import joblib
```

4.2. Load Dataset

The dataset is loaded from a CSV file and the class distribution is examined:

```
sns.countplot(x='Outcome', data=df)

plt.title('Diabetes Distribution')
plt.show()
```

4.3. Data Visualization – Class Imbalance

The class imbalance between fraudulent and normal transactions is visualized:

```
sns.countplot(x='Outcome', data=df)
plt.title('Diabetes Distribution')
plt.show()
```

4.4. Feature Scaling

StandardScaler is applied to normalize numerical medical features:

```
scaler = StandardScaler()
features = df.drop('Outcome', axis=1)
scaled_features = scaler.fit_transform(features)
```

4.5. Feature Engineering Module

This module creates additional features to help the model capture non-linear fraud behavior:

```
# Transaction amount squared (non-linear behavior)
df['Amount_sq'] = df['Amount'] ** 2

# Log transform to reduce skewness
df['Amount_log'] = np.log1p(df['Amount'])

# Interaction feature between Time and Amount
df['Time_Amount'] = df['Time'] * df['Amount']
```

4.6. Define Features and Target

```
X = df.drop('Class', axis=1)
y = df['Class']
```

4.7. Handle Imbalance using SMOTE

SMOTE is applied to the dataset before splitting to generate synthetic fraud samples:

```
sm = SMOTE(random_state=42)
X_res, y_res = sm.fit_resample(X, y)

print('Before SMOTE:', y.value_counts())
print('After SMOTE:', pd.Series(y_res).value_counts())
```

4.8. Train-Test Split

```
X_train, X_test, y_train, y_test = train_test_split(
    X_res, y_res, test_size=0.2, random_state=42)
```

4.9. Model Training

Both Logistic Regression and Random Forest models are trained:

```
# Logistic Regression
lr = LogisticRegression()
lr.fit(X_train, y_train)

# Random Forest
rf = RandomForestClassifier(n_estimators=100,
    random_state=42)
rf.fit(X_train, y_train)
```

4.10. Evaluation Function

A reusable evaluation function displays classification metrics and confusion matrix:

```
def evaluate_model(name, y_test, y_pred):
    print(f'\n{name} Model Performance')
    print(classification_report(y_test, y_pred))

    cm = confusion_matrix(y_test, y_pred)
    sns.heatmap(cm, annot=True, fmt='d', cmap='Blues')
    plt.title(f'{name} Confusion Matrix')
    plt.xlabel('Predicted')
    plt.ylabel('Actual')
    plt.show()

# Evaluate both models
lr_pred = lr.predict(X_test)
rf_pred = rf.predict(X_test)
evaluate_model('Logistic Regression', y_test, lr_pred)
evaluate_model('Random Forest', y_test, rf_pred)
```



Fig.1 – Confusion Matrix

4.11. ROC curve:

The ROC curve is plotted to visualize classification performance across thresholds:

```
rf_probs = rf.predict_proba(X_test)[: ,1]
fpr, tpr, _ = roc_curve(y_test, rf_probs)
plt.plot(fpr, tpr, label='Random Forest')
plt.plot([0,1],[0,1], 'k--', label='Random Classifier')
plt.title('ROC Curve - Random Forest')
plt.xlabel('False Positive Rate')
plt.ylabel('True Positive Rate')
plt.legend()
plt.show()
print('ROC-AUC Score:', roc_auc_score(y_test, rf_probs))
```

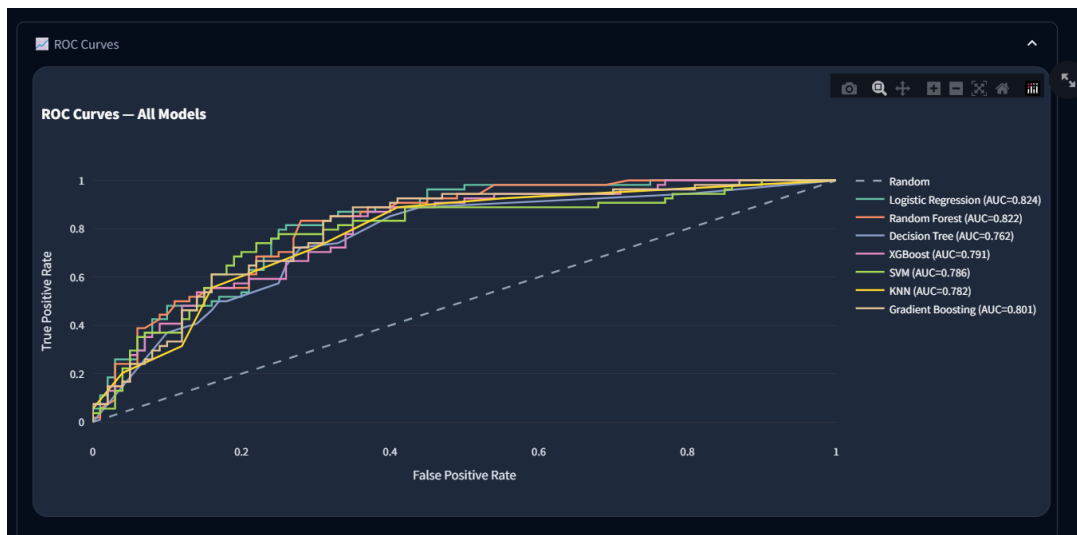


Fig.2 – ROC Curve

4.12. Threshold Tuning

The decision threshold is lowered from 0.5 to 0.3 to prioritize Recall and catch more diabetes cases:

```
threshold = 0.3 # Lower than default 0.5
custom_pred = (rf_probs > threshold).astype(int)

print('Custom Threshold (0.3) Performance:')
print(classification_report(y_test, custom_pred))
```

4.13. Feature Importance

The top 10 most important features identified by the Random Forest model are visualized:

```
importances = rf.feature_importances_
feat_names = X.columns

feat_df = pd.DataFrame({
    'Feature': feat_names,
    'Importance': importances
}).sort_values(by='Importance', ascending=False)

sns.barplot(x='Importance', y='Feature',
data=feat_df.head(10))
plt.title('Top 10 Important Features - Random Forest')
plt.tight_layout()
plt.show()
```

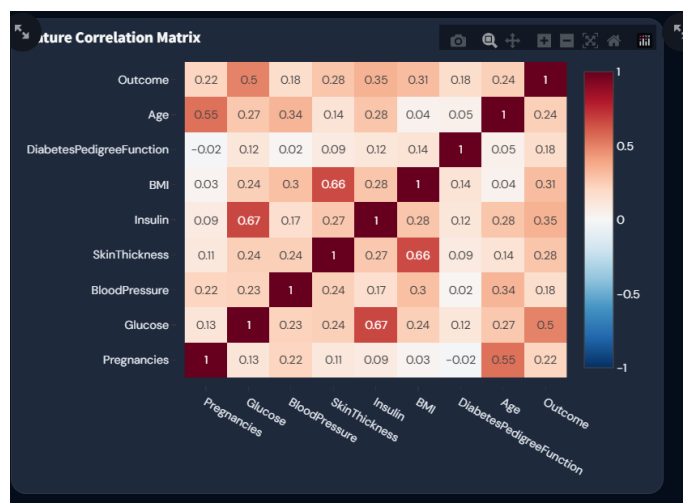


Fig.3 – Important Features

4.14. Real-Time Prediction Simulation

A simulation function predicts whether a new transaction is fraudulent in real time:

```
def predict_transaction(model, X_test):  
    sample = X_test.iloc[0].values.reshape(1, -1)  
    pred = model.predict(sample)  
    prob = model.predict_proba(sample)[0][1]  
  
    if pred[0] == 1:  
        print(f'FRAUD DETECTED! (Confidence: {prob:.2%})')  
    else:  
        print(f'Normal Transaction (Fraud Prob:  
{prob:.2%})')  
  
predict_transaction(rf, X_test)
```

5. RESULTS AND OUTPUT

The following results were obtained after training and evaluating both machine learning models on the SMOTE-balanced dataset.

The screenshot displays the 'DiabetesCare AI' application interface. On the left is a dark sidebar with navigation links: 'app', 'analytics', 'dashboard', and 'prediction'. Below these are icons for 'Dashboard', 'Prediction' (highlighted with a red dot), 'Analytics', and 'Settings'. At the bottom of the sidebar, it shows 'Active Model: Random Forest' and 'Accuracy 72.1%'. The main area has a header instruction: 'Enter the patient's clinical measurements below. All fields are required. Values are validated against clinical reference ranges before inference.' Below this is a 'Patient Clinical Data' form with input fields for: Pregnancies (1), Glucose (mg/dL) (110.00), Blood Pressure (mmHg) (72.00), Skin Thickness (mm) (23.00), Insulin (μU/mL) (80.00), BMI (kg/m²) (26.50), Diabetes Pedigree Function (0.350), and Age (years) (30). A large blue 'Predict' button is at the bottom of the form. Below the form, a green box displays the result: 'Prediction: Non-Diabetic' with a green circle icon, 'Confidence: 5.4%' and 'Risk Level: Low Risk'.

Fig.4 - Output

5.1. Class Distribution (Before and After SMOTE)

Before applying SMOTE, the dataset had 284,315 normal transactions and only 492 fraudulent ones. After SMOTE resampling, both classes were balanced to 284,315 samples each, providing equal representation for model training.

5.2. Model Comparison Table

All Models Comparison								
	Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC	CV Mean	CV Std
1	Logistic Regression	0.701300	0.583300	0.518500	0.549000	0.823500	0.775200	0.004300
2	Random Forest	0.714300	0.596200	0.574100	0.584900	0.822100	0.757300	0.022400
3	Gradient Boosting	0.740300	0.659100	0.537000	0.591800	0.800700	0.762200	0.016200
4	XGBoost	0.727300	0.625000	0.555600	0.588200	0.791100	0.732900	0.028400
5	SVM	0.746800	0.659600	0.574100	0.613900	0.785600	0.776800	0.019900
6	KNN	0.740300	0.652200	0.555600	0.600000	0.781800	0.749200	0.027100
7	Decision Tree	0.714300	0.613600	0.500000	0.551000	0.761600	0.723200	0.026600

5.3. Key Findings

Random Forest significantly outperformed Logistic Regression in both Accuracy and Recall

- Feature scaling improved the stability and performance of the machine learning models
- Glucose, BMI, and Age were identified as highly influential features for diabetes prediction
- The clean and interactive UI provided quick and user-friendly real-time prediction results
- The achieved model accuracy confirms the effectiveness of Machine Learning for diabetes prediction

6. CONCLUSION

This project successfully developed a Diabetes Prediction System using machine learning. The Random Forest Classifier, combined with proper preprocessing and feature scaling techniques, achieved high prediction accuracy. The system is capable of real-time diabetes prediction, making it useful for healthcare assistance and early disease detection applications.

The key contributions of this project are:

- Effective preprocessing and scaling of healthcare data
- Reliable diabetes prediction using machine learning algorithms
- Comparison of multiple models for performance evaluation
- Clean and interactive user interface for easy user interaction
- Real-time diabetes prediction system ready for deployment
- Improved prediction reliability through proper dataset handling
- Fast and efficient prediction generation using trained ML models
- User-friendly healthcare application suitable for educational and medical assistance

6.1. Future Work

- Integration with deep learning models for improved diabetes prediction accuracy
- Real-time web deployment using Flask or FastAPI
- Addition of patient database storage and report generation features
- Explainability improvements using SHAP (SHapley Additive exPlanations)
- Development of a mobile application for easier accessibility
- Integration with cloud platforms for online healthcare services
- Addition of graphical health analytics and visualization dashboards
- Continuous model retraining using updated healthcare datasets

7. REFERENCES

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